

Implications of seasonal variations, host and vector migration on spatial spread of sleeping sickness: Insights from a mathematical model

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ABSTRACT

In this paper, we proposed and analyzed a non-autonomous sleeping sickness that incorporates host and vector movement in a periodic environment. The proposed model is a two patch framework that has three species—humans, animals and tsetse flies. In addition to seasonality and mobility aspects, the proposed model also incorporates vector-bias. Due to seasonal variation in forage and vegetation cover, we assumed that mobility of animals and vectors from one patch to another is assumed to vary seasonally while that of humans is considered to be constant. Analysis of the model has shown that there exists a threshold parameter, commonly known as the reproduction number, which plays a significant role on persistence and extinction of the disease. In particular, we have demonstrated that if the basic reproduction number is less than unity then the disease dies out. However, if the basic reproduction is greater than unity the model admits at least one positive periodic solution. Our results show that seasonality and migration plays an important role in shaping the long term dynamics of sleeping sickness, which subsequently impacts the design of its control strategies.

1. Introduction

Zoonotic diseases, such as rift valley fever, influenza A viruses, anthrax, rabies, Ebola and Human African Trypanosomiasis (HAT), remain a significant public health challenge. HAT also known as sleeping sickness is a vector-borne infection caused by the parasitic protozoa *Trypanosoma brucei gambiense* in West and Central Africa and *Trypanosoma brucei rhodesiense* in East Africa [2]. HAT can be transmitted to humans through over 20 species of *Glossina* tsetse flies [1]. According to the World Health Organization (WHO) [2], HAT threatens millions of people in 36 countries in sub-Saharan Africa. In particular, HAT mainly affects people who reside in remote rural areas with limited access to adequate health services [2].

Due to sustainable control efforts, in 2009, there were 9878 HAT cases, the lowest record in 50 years [2]. Overall, between 2000 and 2012 the number of HAT cases reduced by almost 73% and as a consequence WHO neglected tropical diseases road map set the target of eliminating HAT as a public health problem by 2020 and interruption of transmission (zero cases) for 2030 [3]. Despite these efforts, however, WHO estimates that approximately 65 million people are at risk to HAT infection [2].

For the development of effective disease management programmes, understanding the transmission dynamics of HAT remains imperative. Various tools and techniques can be used to explore the dynamics of communicable and non-communicable diseases. Among them, mathematical modeling has been found to be a successful tool and has been used extensively by researchers and policy makers to gain insights on disease dynamics as well as guiding the development of disease mitigation strategies [4–9]. As of to date, several mathematical models have greatly contributed to shed light on different aspects that govern the dynamics of HAT (e.g. [1,10–21]). Ackley et al. [12] formulated a dynamic sleeping sickness model with a goal to infer on the effect of temperature on tsetse fly mortality rate. Their study demonstrated that tsetse fly mortality strongly depends on temperature. Lord et al. [13] further reaffirmed the results from Ackley et al.'s work that temperature is a key factor in sleeping sickness dynamics. In particular, Lord et al. [13] utilized a mathematical model to demonstrate that there are optimal temperatures that are suitable for the development of the tsetse fly at its different life cycle. Madsen et al. [17] formulated a mathematical framework, to study the effects of seasonality in tsetse fly abundance and trypanosomiasis transmission. Ndeff-mbah et al. [21] formulated a dynamical model to characterize the impact of vector

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migration on the effectiveness of strategies to control gambiense human African trypanosomiasis. Their study suggested that the WHO 2030 goal of gambiense human African trypanosomiasis elimination is, at least in theory, achievable. These studies and several others cited therein improved our understanding on sleeping sickness transmission and control. Despite these efforts, however, several challenges remain in the mathematical modeling of HAT.

Recent statistical studies carried out in Tanzania, by Kimaro et al. [22] and Ngongolo et al. [23] demonstrated that animal dispersal, which depends on seasonal variations, has a strong influence on the dynamics of HAT. To the best of our knowledge, a mathematical model that aims to explore the implications of seasonal variations, host and vector migration on spatial spread of sleeping sickness is yet to be proposed. To that end, in this paper we developed and analyzed a non-autonomous two patch model for sleeping sickness, with a goal to explore the implications of seasonal variations, host and vector migration on spatial spread of the disease. Each patch is assumed to have distinct populations and infection characteristics. Further, the proposed model comprise of humans, animals and tsetse flies. Dispersal of animals and vectors between the two patches is assumed to depend on seasonal changes. This assumption is based on seasonal changes in forage which in-turn leads to animal migration. Studies on tsetse fly habitat preference [24], have revealed that these parasites prefer to live in an area with dense vegetation cover. It is imperative to assume that changes in vegetation cover are influenced by seasonal changes, that in-turn leads to vector migration. According to Hargrove [25], tsetse flies, in particular the Savannah species such as *G. m. morsitans* and *G. pallidipes* can move about 1 km a day and a fly front can advance about 25 km in a year. Apart from seasonal migration of vectors and animals, the proposed model also incorporates vector-bias phenomena. Prior studies on sleeping sickness [1] have revealed that tsetse flies prefer animal blood meals than human blood meals. Herein, we refer to this phenomena as vector-bias.

We organize the paper as follows: In Section 2, we present the methods and results. The results are both analytical and numerical. In particular, we proposed a two-patch sleeping sickness model, computed the reproduction number as well as analyzing the stability of the model steady-states. Numerical simulations were also carried out to support analytical findings. Section 3 presents the discussions of this paper.

2. Methods and results

2.1. Epidemiological model

We developed a two-patch non-autonomous model framework, formally described by a system of ordinary differential equations (ODEs), which allows us to study the of impact host and vector mobility on sleeping sickness transmission and control. Sleeping sickness epidemiology primarily centres on the interactions between the hosts, which can be either human or animal, and the tsetse flies [1]. Thus, the proposed framework incorporates two hosts-human and animals as well as the vector-tsetse flies. We will use the subscript 1 and 2 to denote the two patches. Further, we assume that humans, animals and vectors in patch 1 are capable of migrating to patch 2 and vice-versa. Due to seasonal variation in forage we assume that animal movement between the two patches is periodic [23], however, mobility of humans between the patches is assumed to independent of seasonal variations. Effects of seasonality on the development and habitation preference for tsetse flies is well documented [24] and it is imperative to assume that their dispersal depends on seasonal variability.

For humans, the total population in patch i ($i = 1, 2$) at time t , $N_{Hi}(t)$ constitute: susceptible individuals $S_{Hi}(t)$, incubating patients $E_{Hi}(t)$, and two compartments related to the distinct stages of sleeping sickness, 1st or asymptomatic stage $I_{Hi}(t)$ and the 2nd stage where trypanosomes have reached the cerebro-spinal fluid $R_{Hi}(t)$. Thus,

$$N_{Hi}(t) = S_{Hi}(t) + E_{Hi}(t) + I_{Hi}(t) + R_{Hi}(t). \quad (1)$$

For animals, in patch i at time t we have: susceptible $S_{Ai}(t)$, incubating $E_{Ai}(t)$, infectious $I_{Ai}(t)$ and recovered $R_{Ai}(t)$. As suggested by Stone and Chitnis [15], we assume that animals can recover from infection and are then temporary immuned, R_{Ai} , before returning to the susceptible state. Thus, the total animal population in patch i at time t is

$$N_{Ai}(t) = S_{Ai}(t) + E_{Ai}(t) + I_{Ai}(t) + R_{Ai}(t). \quad (2)$$

Furthermore, the vector population in patch i at time t constitute: susceptible $S_{Vi}(t)$, incubating $E_{Vi}(t)$ and infectious $I_{Vi}(t)$. Thus, the total population of the vectors in patch i at time t is given by

$$N_{Vi}(t) = S_{Vi}(t) + E_{Vi}(t) + I_{Vi}(t). \quad (3)$$

Once infected, vectors are assumed to remain infectious for their entire life span. Additional assumptions that govern the proposed model (5) are as follows:

- (i) Recruitment rate occurs in humans, animals and vectors at b_{Hi} , b_{Ai} and $b_{Vi}(t)$, respectively. Humans, animals and vectors suffer natural mortality at rate μ_{Hi} , μ_{Ai} and $\mu_{Vi}(t)$, respectively. Susceptible host and vector population are assumed to positive at the initial time, that is,

$$S_{Hi}(0) > 0, \quad S_{Ai}(0) > 0, \quad S_{Vi}(0) > 0. \quad (4)$$

- (ii) Both human and animal hosts, susceptibles become infected via an infectious tsetse bite. Tsetse flies have a per capita biting rate $\sigma_{Vi}(t)$. Tsetse flies are known to prefer animal blood meals compared to human blood meals [1], hence we assume that these bites are disproportionately distributed among available animal

Table 1
Description of model parameters of system (5).

Symbol	Description
N_{H1}, N_{H2}	Total human population size.
N_{A1}, N_{A2}	Total animal population size.
N_{V1}, N_{V2}	Total tsetse population size.
η	Relative tsetse preference for animal hosts.
τ	Phase-shifting parameter.
b_{Ai}, b_{Hi}	Birth rate for the hosts.
b_{Vi}^0	Averaged birth rate of the vectors.
μ_{Vi}^0	Averaged mortality rate of the vectors.
μ_{Ai}, μ_{Hi}	Natural mortality rate for the hosts.
κ_{Vi}^0	Average incubation rate for the vectors.
κ_{Ai}, κ_{Hi}	Incubation rate for the hosts.
σ_{Vi}^0	Average vector biting rate.
α_{Ai}, α_{Hi}	Recovery rate of the infectious host.
γ_{Ai}, γ_{Hi}	Immunity waning rate for the recovered host.
β_{VHi}	The probability of transmission of infection from an infectious vector to a susceptible human given that contact between the two occurs.
β_{VAi}	The probability of transmission of infection from an infectious vector to a susceptible animal given that contact between the two occurs.
β_{HVi}	The probability that a vector becomes infected after biting an infectious human.
β_{AVi}	The probability that a vector becomes infected after biting an infectious animal.
m_{ij}	Migration rate of humans from patch j to i .
n_{ij}	Migration rate of clinically infected humans from patch j to i .
p_{ij}^0	Averaged migration rate of animals from patch j to i .
q_{ij}^0	Averaged migration rate of infectious animals from patch j to i .
θ_{ij}^0	Averaged migration rate of vectors from patch j to i .

and human host per patch. Therefore, we assume that tsetse flies have a preference for animal blood meals η times than over human blood. When the bite involves a susceptible host and an infectious tsetse fly, there is a likelihood of disease transmission. The probability of pathogen transmission from an infectious tsetse fly to a susceptible human in patch i is β_{VHi} . Similarly, the probability of disease transmission from an infectious tsetse fly to a susceptible animal in patch i is β_{VAi} . Currently there are no vaccines for sleeping sickness, however, contact between humans and vectors can be minimized through several preventative strategies such as avoiding bushy areas, and wearing long-sleeved garments of medium-weight material in neutral colors that blend with the background environment. Here we assume that individuals in patch i are capable of implementing preventative strategies by a factor u_i , with $0 \leq u_i \leq 1$. Thus, if $u_i \approx 1$ it implies that contact between vectors and human in patch i is highly unlikely and $u_i \approx 0$ means the contribution of preventative strategies to minimize human and vector contact is insignificant.

- (iii) Humans and animals exposed to infection become infectious at a rate κ_{Hi} and κ_{Ai} , respectively. For humans, recovery occurs after successful treatment at rate α_{Hi} . Available evidence suggests that recovered individuals are susceptible to future infections [1,26]. Therefore parameter γ_{Hi} models immunity waning in human population. In contrast, some animals can recover naturally [1,15,27]. Hence we assume that infectious animals recover naturally at rate α_{Ai} and the immunity of recovered animals wanes out at rate γ_{Ai} .
- (iv) Mobility of susceptible, incubating and recovered humans occurs from patch 1 to patch 2 and vice-versa is modeled by m_{12} and m_{21} , respectively. In general, one can define the movement of susceptible, incubating and recovered humans from patch i to patch j and vice-versa by m_{ij} and m_{ji} , respectively, with $i, j = 1, 2$, $i \neq j$. Furthermore, we assume that infectious humans in each patch are less likely to move from one patch to another compared to susceptible, incubating and recovered humans, due to illness. Hence we modeled movement of infectious humans from patch i to patch j and vice-versa by n_{ij} and n_{ji} , respectively, with the assumption that $n_{ij} \leq m_{ij}$.
- (v) A recent study carried out in the Maasai Steppe, Tanzania, suggests that mobility of cattle from one area to another depends on seasonal changes [23], and this can be attributed to seasonal variability in forage and water sources. From the same study carried out in the Maasai Steppe, Tanzania, it was also noted that seasonal cattle movement have an influence on prevalence of trypanosome infections in cattle. Based on this assertion, here, we assume that mobility of animals from patch i to patch j is time dependent. Therefore we modeled the movement of animals from patch i to patch j and vice-versa by rate $p_{ij}(t)$ and $p_{ji}(t)$, respectively. Due to illness we assume that less infectious animals will be capable of moving between the two patches compared to susceptible, exposed and recovered ones. Therefore, parameter $q_{ij}(t)$ and $q_{ji}(t)$ accounts for mobility of infectious animals.
- (vi) Susceptible vectors become infected following a blood meal from an infectious host. The probability of infection transmission from an infectious human to a susceptible vector is modeled by β_{HVi} . Similarly, the probability of infection transmission from an infectious animal to a susceptible vector is β_{AVi} . Upon infection, tsetse flies incubates the pathogen for $1/\kappa_{Vi}(t)$ days, after which they become infectious for their entire life span. Mobility of tsetse flies from patch i to j and vice-versa is modeled by $\theta_{ij}(t)$ and $\theta_{ji}(t)$, respectively.

Based on these assumptions we have the following system of ODEs:

$$\begin{aligned}
 S'_{Hi}(t) &= b_{Hi}N_{Hi} - \frac{\sigma_{Vi}(t)(1-u_i)\beta_{VHi}I_{Vi}(t)S_{Hi}(t)}{N_{Hi}(t) + \eta N_{Ai}(t)} - \mu_{Hi}E_{Hi}(t) - m_{ij}S_{Hj}(t) + m_{ji}S_{Hi}(t) - \kappa_{Hi}E_{Hi}(t) + \gamma_{Hi}R_{Hi}(t) + \gamma_{Ai}R_{Ai}(t) - \theta_{ij}(t)S_{Vj}(t) - \theta_{ji}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) - \theta_{ij}(t)E_{Vj}(t) - \theta_{ji}(t)E_{Vi}(t), \\
 E'_{Hi}(t) &= \frac{\sigma_{Vi}(t)(1-u_i)\beta_{VHi}I_{Vi}(t)S_{Hi}(t)}{N_{Hi}(t) + \eta N_{Ai}(t)} - \mu_{Hi}E_{Hi}(t) - m_{ij}E_{Hj}(t) + m_{ji}E_{Hi}(t) - \kappa_{Hi}E_{Hi}(t) + \gamma_{Hi}R_{Hi}(t) + \gamma_{Ai}R_{Ai}(t) - \theta_{ij}(t)S_{Vj}(t) - \theta_{ji}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) - \theta_{ij}(t)E_{Vj}(t) - \theta_{ji}(t)E_{Vi}(t), \\
 I'_{Hi}(t) &= \kappa_{Hi}E_{Hi}(t) - (\mu_{Hi} + \alpha_{Hi})I_{Hi}(t) + n_{ij}I_{Hj}(t) - n_{ji}I_{Hi}(t) + \gamma_{Ai}R_{Ai}(t) + \theta_{ji}(t)S_{Vj}(t) - \theta_{ij}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) + \theta_{ji}(t)E_{Vj}(t) - \theta_{ij}(t)E_{Vi}(t), \\
 R'_{Hi}(t) &= \alpha_{Hi}I_{Hi}(t) - (\mu_{Hi} + \gamma_{Hi})R_{Hi}(t) + m_{ji}R_{Hj}(t) - m_{ij}R_{Hi}(t) + \gamma_{Ai}R_{Ai}(t) + \theta_{ji}(t)S_{Vj}(t) - \theta_{ij}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) + \theta_{ji}(t)E_{Vj}(t) - \theta_{ij}(t)E_{Vi}(t), \\
 S'_{Ai}(t) &= b_{Ai}N_{Ai} - \frac{\eta\sigma_{Vi}(t)\beta_{VAi}I_{Vi}(t)S_{Ai}(t)}{N_{Hi}(t) + \eta N_{Ai}(t)} - \mu_{Ai}S_{Ai}(t) + p_{ji}(t)S_{Aj}(t) - p_{ij}(t)S_{Ai}(t) + \gamma_{Ai}R_{Ai}(t) + \theta_{ji}(t)S_{Vj}(t) - \theta_{ij}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) + \theta_{ji}(t)E_{Vj}(t) - \theta_{ij}(t)E_{Vi}(t), \\
 E'_{Ai}(t) &= \frac{\eta\sigma_{Vi}(t)\beta_{VAi}I_{Vi}(t)S_{Ai}(t)}{N_{Hi}(t) + \eta N_{Ai}(t)} - (\mu_{Ai} + \kappa_{Ai})E_{Ai}(t) + p_{ji}(t)E_{Aj}(t) - p_{ij}(t)E_{Ai}(t) + \theta_{ji}(t)S_{Vj}(t) - \theta_{ij}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) + \theta_{ji}(t)E_{Vj}(t) - \theta_{ij}(t)E_{Vi}(t), \\
 I'_{Ai}(t) &= \kappa_{Ai}E_{Ai}(t) - (\mu_{Ai} + \alpha_{Ai})I_{Ai}(t) + q_{ji}(t)I_{Aj}(t) - q_{ij}(t)I_{Ai}(t) + \theta_{ji}(t)S_{Vj}(t) - \theta_{ij}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) + \theta_{ji}(t)E_{Vj}(t) - \theta_{ij}(t)E_{Vi}(t), \\
 R'_{Ai}(t) &= \alpha_{Ai}I_{Ai}(t) - (\mu_{Ai} + \gamma_{Ai})R_{Ai}(t) + p_{ji}(t)R_{Aj}(t) - p_{ij}(t)R_{Ai}(t) + \theta_{ji}(t)S_{Vj}(t) - \theta_{ij}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) + \theta_{ji}(t)E_{Vj}(t) - \theta_{ij}(t)E_{Vi}(t), \\
 S'_{Vi}(t) &= b_{Vi}(t)N_{Vi} - \frac{\sigma_{Vi}(t)\beta_{HVi}I_{Hi}(t)S_{Vi}(t)}{N_{Hi}(t) + \eta N_{Ai}(t)} - \frac{\eta\sigma_{Vi}(t)\beta_{AVi}I_{Ai}(t)S_{Vi}(t)}{N_{Hi}(t) + \eta N_{Ai}(t)} - \mu_{Vi}(t)S_{Vi}(t) + \theta_{ji}(t)S_{Vj}(t) - \theta_{ij}(t)S_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) + \theta_{ji}(t)E_{Vj}(t) - \theta_{ij}(t)E_{Vi}(t), \\
 E'_{Vi}(t) &= \frac{\sigma_{Vi}(t)\beta_{HVi}I_{Hi}(t)S_{Vi}(t)}{N_{Hi}(t) + \eta N_{Ai}(t)} + \frac{\eta\sigma_{Vi}(t)\beta_{AVi}I_{Ai}(t)S_{Vi}(t)}{N_{Hi}(t) + \eta N_{Ai}(t)} - \kappa_{Vi}(t)E_{Vi}(t) - \mu_{Vi}(t)E_{Vi}(t) - \theta_{ij}(t)E_{Vj}(t) - \theta_{ji}(t)E_{Vi}(t), \\
 I'_{Vi}(t) &= \kappa_{Vi}(t)E_{Vi}(t) - \mu_{Vi}(t)I_{Vi}(t) + \theta_{ji}(t)I_{Vj}(t) - \theta_{ij}(t)I_{Vi}(t).
 \end{aligned}
 \tag{5}$$

Remark 2.1.

Model parameters that are functions of time accounts for the effects seasonal variations.

We now provide comprehensive expressions to model parameters that are meant to capture seasonal fluctuations, that is;

$$\left. \begin{aligned} \sigma_{Vi}(t) &= \sigma_{Vi}^0 [1 + \sigma_{Vi}^1 \cos(\omega t + \tau)], \\ \kappa_{Vi}(t) &= \kappa_{Vi}^0 [1 + \kappa_{Vi}^1 \cos(\omega t + \tau)], \\ b_{Vi}(t) &= b_{Vi}^0 [1 + b_{Vi}^1 \cos(\omega t + \tau)], \\ \mu_{Vi}(t) &= \mu_{Vi}^0 [1 + \mu_{Vi}^1 \cos(\omega t + \tau)], \\ p_{ij}(t) &= p_{ij}^0 [1 + p_{ij}^1 \cos(\omega t + \tau)], \\ q_{ij}(t) &= q_{ij}^0 [1 + q_{ij}^1 \cos(\omega t + \tau)], \end{aligned} \right\} \quad (6)$$

where ω and τ represent the period and phase, respectively. Furthermore, the positive constants $\sigma_{Vi}^0, \kappa_{Vi}^0, b_{Vi}^0, \mu_{Vi}^0, p_{ij}^0, q_{ij}^0$ represents the base-line values or the times averages of parameters $\sigma_{Vi}(t), \kappa_{Vi}(t), b_{Vi}(t), \mu_{Vi}(t), p_{ij}$ and $q_{ij}(t)$, respectively. In addition, nonnegative constants $\sigma_{Vi}^1, \kappa_{Vi}^1, b_{Vi}^1, \mu_{Vi}^1, p_{ij}^1$ and q_{ij}^1 denotes the amplitude of oscillations in $\sigma_{Vi}(t), \kappa_{Vi}(t), b_{Vi}(t), \mu_{Vi}(t), p_{ij}$ and $q_{ij}(t)$, respectively. Table 1 presents a summary of the model parameters and their definitions.

2.2. The reproduction number

In the absence of sleeping sickness in the community, that is, $E_{Hi} = I_{Hi} = R_{Hi} = 0, E_{Ai} = I_{Ai} = R_{Ai} = 0, E_{Vi} = I_{Vi} = 0$, system reduces to

$$\left\{ \begin{aligned} S'_{H1}(t) &= b_{H1}S_{H1} - (\mu_{H1} + m_{12})S_{H1} + m_{21}S_{H2}, \\ S'_{H2}(t) &= b_{H2}S_{H2} - (\mu_{H2} + m_{21})S_{H2} + m_{12}S_{H1}, \\ S'_{A1}(t) &= b_{A1}S_{A1} - (p_{12}(t) + \mu_{A1})S_{A1} + p_{21}S_{A2}, \\ S'_{A2}(t) &= b_{A2}S_{A2} - (p_{21}(t) + \mu_{A2})S_{A2} + p_{12}S_{A1}, \\ S'_{V1}(t) &= b_{V1}(t)S_{V1}(t) - (\theta_{12}(t) + \mu_{V1}(t))S_{V1} \\ &\quad + \theta_{21}(t)S_{V2}, \\ S'_{V2}(t) &= b_{V2}(t)S_{V2}(t) - (\theta_{21}(t) + \mu_{V2}(t))S_{V2} \\ &\quad + \theta_{12}(t)S_{V1}, \end{aligned} \right\} \quad (7)$$

Solving (7) leads to $S_{Hi}^0 = N_{Hi}, S_{Ai}^0 = N_{Ai}$ and $S_{Vi} = N_{Vi}$. Let $N_H(t) = \sum_{i=1}^2 N_{Hi}, N_A(t) = \sum_{i=1}^2 N_{Ai}$, and $N_V(t) = \sum_{i=1}^2 N_{Vi}$, then one can easily verify that the domain of biological interest is positively invariant and attracting all orbits with respect to the model (5).

$$\Omega = \left\{ \begin{pmatrix} N_H(t) \\ N_A(t) \\ N_V(t) \end{pmatrix} \in \mathbb{R}_+^{22} \mid \begin{aligned} N_H(t) &\leq N_{H1} + N_{H2}, \\ N_A(t) &\leq N_{A1} + N_{A2}, \\ N_V(t) &\leq N_{V1} + N_{V2}, \end{aligned} \right\},$$

By the definition of Bacaër and Guernaoui [28] and the general calculation procedure in Wang and Zhao [29]. Utilizing the next-generation method, the following matrix represents the generation of new infections $F(t)$ (evaluated at the disease-free equilibrium):

$$F(t) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & f_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & f_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & f_3 & 0 & f_4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (8)$$

with,

$$f_1 = \frac{\sigma_{V1}(t)(1 - u_1)\beta_{HV1}N_{H1}}{N_{H1} + \eta N_{A1}}, f_2 = \frac{\eta\sigma_{V1}(t)\beta_{VA1}N_{A1}}{N_{H1} + \eta N_{A1}},$$

$$f_3 = \frac{\sigma_{V1}(t)(1 - u_1)\beta_{HV1}N_{V1}}{N_{H1} + \eta N_{A1}}, f_4 = \frac{\eta\sigma_{V1}(t)\beta_{AV1}N_{V1}}{N_{H1} + \eta N_{A1}},$$

$$f_5 = \frac{\sigma_{V2}(t)(1 - u_2)\beta_{VH1}N_{H2}}{N_{H2} + \eta N_{A2}}, f_6 = \frac{\eta\sigma_{V2}(t)\beta_{VA2}N_{A2}}{N_{H2} + \eta N_{A2}},$$

$$f_7 = \frac{\sigma_{V2}(t)(1 - u_2)\beta_{HV2}N_{V2}}{N_{H2} + \eta N_{A2}}, f_8 = \frac{\eta\sigma_{V2}(t)\beta_{AV2}N_{V2}}{N_{H2} + \eta N_{A2}}.$$

Furthermore, from the next-generation matrix method we define the matrix of transfer terms (evaluated at the disease-free equilibrium) as follows:

$$V(t) = \begin{bmatrix} V_{11}(t) & V_{12}(t) \\ V_{21}(t) & V_{22}(t) \end{bmatrix} \quad (9)$$

where

$$V_{11}(t) = \begin{bmatrix} z_1 & 0 & 0 & 0 & 0 & 0 \\ -\kappa_{H1} & z_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & z_3 & 0 & 0 & 0 \\ 0 & 0 & -\kappa_{A1} & z_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & z_5 & 0 \\ 0 & 0 & 0 & 0 & -\kappa_{V1}(t) & z_6 \end{bmatrix},$$

with;

$$\left. \begin{aligned} z_1 &= \mu_{H1} + \kappa_{H1} + m_{12}, \\ z_2 &= \mu_{H1} + \alpha_{H1} + n_{12}, \\ z_3 &= \mu_{A1} + \kappa_{A1} + p_{12}(t), \\ z_4 &= \mu_{A1} + \alpha_{A1} + q_{12}(t), \\ z_5 &= \mu_{V1}(t) + \kappa_{V1}(t) + \theta_{12}(t), \\ z_6 &= \mu_{V1}(t) + \theta_{12}(t), \end{aligned} \right\}$$

$$V_{12}(t) = \begin{bmatrix} -m_{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & -n_{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & -p_{21}(t) & 0 & 0 & 0 \\ 0 & 0 & 0 & -q_{12}(t) & 0 & 0 \\ 0 & 0 & 0 & 0 & -\theta_{21}(t) & 0 \\ 0 & 0 & 0 & 0 & 0 & -\theta_{21}(t) \end{bmatrix},$$

$$V_{21}(t) = \begin{bmatrix} -m_{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & -n_{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & -p_{12}(t) & 0 & 0 & 0 \\ 0 & 0 & 0 & -q_{12}(t) & 0 & 0 \\ 0 & 0 & 0 & 0 & -\theta_{12}(t) & 0 \\ 0 & 0 & 0 & 0 & 0 & -\theta_{12}(t) \end{bmatrix},$$

$$V_{22}(t) = \begin{bmatrix} \xi_1 & 0 & 0 & 0 & 0 & 0 \\ -\kappa_{H2} & \xi_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \xi_3 & 0 & 0 & 0 \\ 0 & 0 & -\kappa_{A2} & \xi_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & \xi_5 & 0 \\ 0 & 0 & 0 & 0 & -\kappa_{V2}(t) & \xi_6 \end{bmatrix},$$

with

$$\left. \begin{aligned} \xi_1 &= \mu_{H2} + \kappa_{H2} + m_{21}, \\ \xi_2 &= \mu_{H2} + \alpha_{H2} + n_{21}, \\ \xi_3 &= \mu_{A2} + \kappa_{A2} + p_{21}(t), \\ \xi_4 &= \mu_{A2} + \alpha_{A2} + q_{21}(t), \\ \xi_5 &= \mu_{V2}(t) + \kappa_{V2}(t) + \theta_{21}(t), \\ \xi_6 &= \mu_{V2}(t) + \theta_{21}(t). \end{aligned} \right\}$$

The basic reproduction number of system (5) in the absence of seasonal forcing is given by

$$[\mathcal{R}_0] = \max\{\mathcal{R}_{01}, \mathcal{R}_{02}\}, \quad (10)$$

where

$$\mathcal{R}_{0i} = \sqrt{\mathcal{R}_{Vi}(\mathcal{R}_{Hi} + \mathcal{R}_{Ai})}$$

with

$$\begin{aligned}\mathcal{R}_{Vi} &= \left(\frac{\kappa_{Vi}^0}{(\mu_{Vi}^0 + \kappa_{Vi}^0 + \theta_{ij}^0)(\mu_{Vi}^0 + \theta_{ij}^0)} \right), \\ \mathcal{R}_{Hi} &= \left(\frac{\kappa_{Hi}}{(\mu_{Hi} + \kappa_{Hi} + m_{ij})(\mu_{Hi} + \alpha_{Hi} + n_{ij})} \right) \times \left(\frac{\sigma_{Vi}^0(1 - u_i)\beta_{HVi}N_{Vi}}{N_{Hi} + \eta N_{Ai}} \right) \\ &\times \left(\frac{\sigma_{Vi}^0(1 - u_i)\beta_{VHi}N_{Hi}}{N_{Hi} + \eta N_{Ai}} \right), \mathcal{R}_{Ai} = \left(\frac{\kappa_{Ai}}{(\mu_{Ai} + \kappa_{Ai} + p_{ij}^0)(\mu_{Ai} + \alpha_{Ai} + q_{ij}^0)} \right) \\ &\times \left(\frac{\eta\sigma_{Vi}^0\beta_{VAi}N_{Ai}}{N_{Hi} + \eta N_{Ai}} \right) \left(\frac{\eta\sigma_{Vi}^0\beta_{AVi}N_{Vi}}{N_{Hi} + \eta N_{Ai}} \right).\end{aligned}$$

with $i \neq j = 1, 2$.

Let $Y(t, s), t \geq s$, be the evolution operator of the linear ω -periodic system. Consider

$$\frac{dy}{dt} = -V(t)y. \quad (11)$$

That is, for each $s \in \mathbb{R}$, the 12×12 matrix $Y(t, s)$ satisfies

$$\frac{dY(t, s)}{dt} = -V(t)Y(t, s), \forall t \geq s, Y(s, s) = I_{12 \times 12}, \quad (12)$$

where $I_{12 \times 12}$ is the 12×12 identity matrix. Thus, the monodromy matrix $\Phi_{-V(t)}$ of (12) equals $Y(t, 0)$, $t \geq 0$. By closely following the approach in Wang and Zhao [29], we let $\varphi(s)$ be ω -periodic in s and the initial distribution of infectious species. Hence $F(s)\varphi(s)$ represent the rate of new infections generated by the infectious species who are introduced in a wholly susceptible population at time s . Thus, for $t \geq s$, $Y(t, s)F(s)\varphi(s)$ accounts for the distribution of the new infections produced by $\varphi(s)$ and remain in the infected compartments at time t . In summary, Wang and Zhao [29] introduced the next-infection operator L for a model in periodic environments as follows:

$$(L\varphi)(t) = \int_0^\infty Y(t, t-s)F(t-s)\varphi(t-s)ds, \quad (13)$$

where $Y(t, s), t \geq s$, is the evolution operator of the linear ω -periodic system $dy/dt = -V(t)y$ and $\varphi(t)$, the initial distribution of infectious individuals, is ω -periodic and nonnegative. The basic reproduction number is then defined as the spectral radius of the next-infection operator,

$$R_0 = \rho(L). \quad (14)$$

For model (5), the evolution operator can be determined by solving the system of differential equations $dy/dt = -V(t)y$ with the initial condition $Y(s, s) = I_{12 \times 12}$. Due to the complex nature of $V(t)$ (15) we will not compute the evolution operator. It is worth noting that the basic reproduction number for periodic models (5) can be evaluated using various numerical approach, for example, one can consider the method described in Ref. [30].

Lemma 1. (Theorem 2.2 in Wang and Zhao [29]). Suppose that $y(t) = (E_{Hi}, I_{Hi}, E_{Ai}, I_{Ai}, E_{Vi}, I_{Vi})$, $i = 1, 2$, represent all the classes of all infected species for the model system (5), such that the its linearization at disease-free equilibrium $\mathcal{E}^0$ is

$$\dot{y}(t) = (F(t) - V(t))y(t), \quad (15)$$

where $F(t)$ and $V(t)$ are defined on (8) and (9), respectively. Furthermore, let $\Psi_{F(t)-V(t)}$ and $\rho(\Psi_{F-V}(\omega))$ be the monodromy matrix of system (15) and the spectral radius of $\Psi_{F-V}(\omega)$, respectively, then we have following statements: (i) $\mathcal{R}_0 = 1$, if and only if $\rho(\Psi_{F-V}(\omega)) = 1$; (ii) $\mathcal{R}_0 > 1$, if and only if $\rho(\Psi_{F-V}(\omega)) > 1$; (iii) $\mathcal{R}_0 < 1$, if and only if $\rho(\Psi_{F-V}(\omega)) < 1$. Therefore, the

disease-free equilibrium $\mathcal{E}^0$ of the model system (5) is locally asymptotically stable if $\mathcal{R}_0 < 1$ and unstable if $\mathcal{R}_0 > 1$.

Now we introduce the linear ω -periodic system

$$\frac{d\omega}{dt} = \left(-V(t) + \frac{F(t)}{\lambda} \right) \omega, t \in \mathbb{R}_+ \quad (16)$$

with parameter $\lambda \in \mathbb{R}$. Let $W(t, s, \lambda), t \geq s$, be the evolution operator of system (5) on $\mathbb{R}^{22}$. It follows that $\Phi_{F-V}(t) = W(t, 0, 1), t \geq 0$. Hence, one can conclude that

$$\Phi_{F/\lambda-V}(t) = W(t, 0, \lambda). \quad (17)$$

Based on arguments and general computation algorithm of the basic reproduction number in Wang and Zhao [29], it follows that basic reproduction number of model system (5) $\mathcal{R}_0$ is the unique solution of $r(W(\omega, 0, \lambda)) = 1$.

In what follows, we now demonstrate that the reproduction number $\mathcal{R}_{0i}$ is an important threshold parameter for disease extinction and persistence. In particular, we have to show that when $\mathcal{R}_0 < 1$, the model system (5) admits a globally asymptotically stable disease-free equilibrium $\mathcal{E}^0$, and if $\mathcal{R}_0 > 1$, the disease persists. The mathematical analysis follows the approach in [31].

Theorem 1. If $\mathcal{R}_0 < 1$ with then the disease-free equilibrium $\mathcal{E}^0$ of system (5) is globally asymptotically stable in Ω .

Proof. According to Lemma 1, if $\mathcal{R}_0 < 1$, then the disease-free equilibrium $\mathcal{E}^0$ of the model system (5) is locally asymptotically stable. Therefore, we demonstrate that for $\mathcal{R}_0 < 1$, the disease-free equilibrium is the global attractor. Assume that $\mathcal{R}_0 < 1$, again from Lemma 1, we have, $\rho(\Psi_{F-V}(\omega)) < 1$. By considering the infected classes we have the following outcome (for $i = 1, 2$):

$$\left. \begin{aligned} \dot{E}_{Hi}(t) &\leq \left(\frac{\sigma_{Vi}(t)(1 - u_i)\beta_{VHi}I_{Vi}}{N_{Hi} + \eta N_{Ai}} \right) S_{Hi}^0 \\ &\quad - (\mu_{Hi} + \kappa_{Hi})E_{Hi} \\ &\quad + m_{ji}E_{Hj} - m_{ij}E_{Hi}, \\ \dot{I}_{Hi}(t) &= \kappa_{Hi}E_{Hi} - (\mu_{Hi} + \alpha_{Hi})I_{Hi} \\ &\quad + n_{ji}I_{Hj} - n_{ij}I_{Hi}, \\ \dot{E}_{Ai}(t) &\leq \left(\frac{\eta\sigma_{Vi}(t)\beta_{VAi}I_{Vi}}{N_{Hi} + \eta N_{Ai}} \right) S_{Ai}^0 \\ &\quad - (\mu_{Ai} + \kappa_{Ai})E_{Ai} \\ &\quad + p_{ji}(t)E_{Aj} - p_{ij}(t)E_{Ai}, \end{aligned} \right\} \quad (18)$$

$$\left. \begin{aligned} \dot{I}_{Ai}(t) &= \kappa_{Ai}E_{Ai} - (\mu_{Ai} + \alpha_{Ai})I_{Ai} \\ &\quad + q_{ji}(t)I_{Aj} - q_{ij}(t)I_{Ai}, \\ \dot{E}_{Vi}(t) &\leq \left(\frac{\sigma_{Vi}(t)(1 - u_i)\beta_{HVi}I_{Hi}}{N_{Hi} + \eta N_{Ai}} \right) S_{Vi}^0 \\ &\quad + \left(\frac{\eta\sigma_{Vi}(t)\beta_{AVi}I_{Ai}}{N_{Hi} + \eta N_{Ai}} \right) S_{Vi}^0 \\ &\quad + \theta_{ji}(t)E_{Vj} - \theta(t)_{ij}E_{Vi}, \\ \dot{I}_{Vi}(t) &= \kappa_{Vi}(t)E_{Vi} - \mu_{Vi}(t)I_{Vi} \\ &\quad + \theta_{ji}(t)I_{Vj} - \theta(t)_{ij}I_{Vi}, \end{aligned} \right\} \quad (19)$$

for $t \geq 0$. Consider the following auxiliary system:

$$\left. \begin{aligned}
\dot{\tilde{E}}_{Hi}(t) &\leq \left(\frac{\sigma_{Vi}(t)(1-u_i)\beta_{VHi}\tilde{I}_{Vi}}{N_{Hi} + \eta N_{Ai}} \right) S_{Hi}^0 - (\mu_{Hi} + \kappa_{Hi})\tilde{E}_{Hi} + m_{ji}\tilde{E}_{Hj} - m_{ij}\tilde{E}_{Hi} + p_{ji}(t)\tilde{E}_{Aj} - p_{ij}(t)\tilde{E}_{Ai} + \theta_{ji}(t)\tilde{E}_{Vj} - \theta(t)_{ij}\tilde{E}_{Vi}, \\
\dot{\tilde{I}}_{Hi}(t) &= \kappa_{Hi}\tilde{E}_{Hi} - (\mu_{Hi} + \alpha_{Hi})\tilde{I}_{Hi} + n_{ji}\tilde{I}_{Hj} - n_{ij}\tilde{I}_{Hi} + p_{ji}(t)\tilde{E}_{Aj} - p_{ij}(t)\tilde{E}_{Ai} + \theta_{ji}(t)\tilde{E}_{Vj} - \theta(t)_{ij}\tilde{E}_{Vi}, \\
\dot{\tilde{E}}_{Ai}(t) &\leq \left(\frac{\eta\sigma_{Vi}(t)\beta_{VAi}\tilde{I}_{Vi}}{N_{Hi} + \eta N_{Ai}} \right) S_{Ai}^0 - (\mu_{Ai} + \kappa_{Ai})\tilde{E}_{Ai} + p_{ji}(t)\tilde{E}_{Aj} - p_{ij}(t)\tilde{E}_{Ai} + \theta_{ji}(t)\tilde{E}_{Vj} - \theta(t)_{ij}\tilde{E}_{Vi}, \\
\dot{\tilde{I}}_{Ai}(t) &= \kappa_{Ai}\tilde{E}_{Ai} - (\mu_{Ai} + \alpha_{Ai})\tilde{I}_{Ai} + q_{ji}(t)\tilde{I}_{Aj} - q_{ij}(t)\tilde{I}_{Ai} + \theta_{ji}(t)\tilde{E}_{Vj} - \theta(t)_{ij}\tilde{E}_{Vi}, \\
\dot{\tilde{E}}_{Vi}(t) &\leq \left(\frac{\sigma_{Vi}(t)(1-u_i)\beta_{VHi}\tilde{I}_{Hi}}{N_{Hi} + \eta N_{Ai}} \right) S_{Vi}^0 + \left(\frac{\eta\sigma_{Vi}(t)\beta_{VAi}\tilde{I}_{Ai}}{N_{Hi} + \eta N_{Ai}} \right) S_{Vi}^0 + \theta_{ji}(t)\tilde{E}_{Vj} - \theta(t)_{ij}\tilde{E}_{Vi}, \\
\dot{\tilde{I}}_{Vi}(t) &= \kappa_{Vi}(t)\tilde{E}_{Vi} - \mu_{Vi}(t)\tilde{I}_{Vi} + \theta_{ji}(t)\tilde{I}_{Vj} - \theta(t)_{ij}\tilde{I}_{Vi},
\end{aligned} \right\} \quad (20)$$

By Lemma 1 and the standard comparison principle, there exist a positive ω -periodic function $\tilde{y}(t)$ such that $y(t) \leq \tilde{y}(t)e^{pt}$, where $\tilde{y}(t) = (\tilde{E}_{Hi}, \tilde{I}_{Hi}, \tilde{E}_{Ai}, \tilde{I}_{Ai}(t), \tilde{E}_{Vi}(t), \tilde{I}_{Vi}(t))$, for $i = 1, 2$, and $p = \frac{1}{\omega} \ln \rho(\psi_{(F-V)(\cdot)}(\omega)) < 0$. Thus we conclude that $y(t) \rightarrow 0$ as $t \rightarrow \infty$, that is,

$$\lim_{t \rightarrow \infty} E_{Hi}(t) = 0, \quad \lim_{t \rightarrow \infty} I_{Hi}(t) = 0, \quad \lim_{t \rightarrow \infty} E_{Ai}(t) = 0,$$

$$\lim_{t \rightarrow \infty} I_{Ai}(t) = 0, \quad \lim_{t \rightarrow \infty} E_{Vi}(t) = 0, \quad \lim_{t \rightarrow \infty} I_{Vi}(t) = 0.$$

Hence it follows that

$$\left. \begin{aligned}
\lim_{t \rightarrow \infty} S_{Hi}(t) &= S_{Hi}^0, & \lim_{t \rightarrow \infty} S_{Ai}(t) &= S_{Ai}^0, \\
\lim_{t \rightarrow \infty} S_{Vi}(t) &= S_{Vi}^0, & \lim_{t \rightarrow \infty} R_{Hi}(t) &= 0, \\
\lim_{t \rightarrow \infty} R_{Ai}(t) &= 0.
\end{aligned} \right\} \quad (21)$$

Thus, we conclude that the disease-free equilibrium $\mathcal{E}^0$ of system (5) is globally asymptotically stable.

Theorem 2. If $R_0 > 1$, then system (5) is uniformly persistent, i.e., there exists a positive constant χ , such that for all initial values of $(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \in \mathbb{R}_+^5 \times \text{Int}(\mathbb{R}_+)^6$, ($i = 1, 2$) the solution of model (5) satisfies:

$$\left. \begin{aligned}
\lim_{t \rightarrow \infty} \inf S_{Hi}(t) &\geq \chi, & \lim_{t \rightarrow \infty} \inf E_{Hi}(t) &\geq \chi, \\
\lim_{t \rightarrow \infty} \inf I_{Hi}(t) &\geq \chi, & \lim_{t \rightarrow \infty} \inf R_{Hi}(t) &\geq \chi, \\
\lim_{t \rightarrow \infty} \inf S_{Ai}(t) &\geq \chi, & \lim_{t \rightarrow \infty} \inf E_{Ai}(t) &\geq \chi, \\
\lim_{t \rightarrow \infty} \inf I_{Ai}(t) &\geq \chi, & \lim_{t \rightarrow \infty} \inf R_{Ai}(t) &\geq \chi, \\
\lim_{t \rightarrow \infty} \inf S_{Vi}(t) &\geq \chi, & \lim_{t \rightarrow \infty} \inf E_{Vi}(t) &\geq \chi, \\
\lim_{t \rightarrow \infty} \inf I_{Vi}(t) &\geq \chi.
\end{aligned} \right\} \quad (22)$$

Proof. Let us define that is, $Y = (S_{Hi}, E_{Hi}, I_{Hi}, R_{Hi}, S_{Ai}, E_{Ai}, I_{Ai}, R_{Ai}, S_{Vi}, E_{Vi}, I_{Vi})$. Let $P : Y \rightarrow Y$ be the Poincaré map associated with our model (5) such that $P(y_0) = u(\omega, y_0) \forall y_0 \in Y$, where $u(t, y_0)$ denotes the unique solution of the system with $u(0, y_0) = y_0$. Furthermore, let us define

$$\begin{aligned}
M_\partial &= \{ (S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), \\
&\quad I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \in \partial Y_0 \\
&\quad P^m(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), \\
&\quad E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \in \partial Y_0, \forall m \geq 0 \}
\end{aligned} \quad (23)$$

and

$$\begin{aligned}
\tilde{M}_\partial &= \left\{ (S_{Hi}, 0, 0, R_{Hi}, S_{Ai}, 0, 0, R_{Ai}, S_{Vi}(0), 0, 0) : \right. \\
&\quad \left. S_{Hi} \geq 0, R_{Hi} \geq 0, S_{Ai} \geq 0, R_{Ai} \geq 0, S_{Vi} \geq 0 \right\}
\end{aligned}$$

We begin by demonstrating that

$$M_\partial = \tilde{M}_\partial. \quad (24)$$

Consider any initial values

$$(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \in \partial Y_0 \setminus \tilde{M}_\partial \quad (25)$$

From system (5) it follows that if $E_{Hi}(0) = 0, I_{Hi}(0) = R_{Hi} = 0, E_{Ai}(0) = 0, I_{Ai}(0) = 0, R_{Ai}(0) = 0, E_{Vi}(0) = I_{Vi}(0) > 0$ then based on assumption (4) we have $E'_{Hi} > 0$ and $E'_{Ai} > 0$. Similarly, if $E_{Hi}(0) = 0, I_{Hi}(0) > 0, R_{Hi} = 0, E_{Ai}(0) = 0, I_{Ai}(0) = 0, R_{Ai}(0) = 0$, and $I_{Vi}(0) = 0$, we have $E_{Vi} > 0$. Thus, we have $(S_{Hi}(t), E_{Hi}(t), I_{Hi}(t), R_{Hi}(t), S_{Ai}(t), E_{Ai}(t), I_{Ai}(t), R_{Ai}(t), S_{Vi}(t), E_{Vi}(t), I_{Vi}(t)) \notin \partial Y_0$ for $0 < t \leq 1$. By the positive invariance of Y_0 , we know that $P^m(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \notin \partial X_0$ for $m \geq 1$, hence $(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \notin M_\partial$, and thus (24) holds.

Now consider the fixed point $M_0 = (S_{Hi}^0, 0, 0, 0, S_{Ai}^0, 0, 0, 0, S_{Vi}^0, 0, 0)$ of the Poincaré map P , where and define $W^s(M_0) = \{y_0 : P^m(y_0) \rightarrow M_0, m \rightarrow \infty\}$. We show that

$$W^s(M_0) \cap Y_0 = \emptyset. \quad (26)$$

Based on the continuity of solutions with respect to the initial conditions, for any $\varepsilon > 0$, there exists $\delta > 0$ small enough such that for all $(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0))$ with $\|(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) - M_0\| \leq \delta$, we have

$$\|u(t, (S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0))) - u(t, M_0)\| < \varepsilon, \quad \forall t \in [0, \omega]. \quad (27)$$

To obtain (26), we claim that

$$\begin{aligned}
&\lim_{m \rightarrow \infty} \sup \|P^m \left(S_{Hi} \left(0 \right), E_{Hi} \left(0 \right), I_{Hi} \left(0 \right), R_{Hi} \left(0 \right), S_{Ai} \left(0 \right), \right. \\
&\quad \left. E_{Ai} \left(0 \right), I_{Ai} \left(0 \right), R_{Ai} \left(0 \right), S_{Vi} \left(0 \right), E_{Vi} \left(0 \right), I_{Vi} \left(0 \right) \right)
\end{aligned}$$

$$-M_0| \geq \delta,$$

$$\forall (S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0),$$

$$E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \in Y_0$$

We prove this claim by contradiction; that is, we suppose

$$\limsup_{m \rightarrow \infty} |P^m \left(S_{Hi} \left(0 \right), E_{Hi} \left(0 \right), I_{Hi} \left(0 \right), R_{Hi} \left(0 \right), S_{Ai} \left(0 \right), \right.$$

$$E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \left. \right)$$

$$-M_0| \geq \delta,$$

$$\forall (S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0),$$

for any $t \geq 0$. Let

$$\begin{aligned} (S_{Hi}(t), E_{Hi}(t), I_{Hi}(t), R_{Hi}(t), S_{Ai}(t), E_{Ai}(t), I_{Ai}(t), \\ R_{Ai}(t), S_{Vi}(t), E_{Vi}(t), I_{Vi}(t)) = u(t, (S_{Hi}(0), E_{Hi}(0), \\ I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), \\ E_{Vi}(0), I_{Vi}(0))) \end{aligned} \quad (31)$$

It follows that

$$\begin{aligned} S_{Hi}^0 - \varepsilon < S_{Hi}(t) < S_{Hi}^0 + \varepsilon, \quad 0 < E_{Hi}(t) < \varepsilon, \\ 0 < I_{Hi}(t) < \varepsilon, \quad 0 < R_{Hi}(t) < \varepsilon, \\ S_{Ai}^0 - \varepsilon < S_{Ai}(t) < S_{Ai}^0 + \varepsilon, \quad 0 < E_{Ai}(t) < \varepsilon, \\ 0 < I_{Ai}(t) < \varepsilon, \quad 0 < R_{Ai}(t) < \varepsilon, \\ S_{Vi}^0 - \varepsilon < S_{Vi}(t) < S_{Vi}^0 + \varepsilon, \quad 0 < E_{Vi}(t) < \varepsilon, \\ 0 < I_{Vi}(t) < \varepsilon. \end{aligned} \quad (32)$$

Then we have

$$\begin{aligned} E'_{Hi}(t) &= \frac{\sigma_{Vi}(t)(1-u_i)\beta_{VHi}I_{Vi}S_{Hi}}{N_{Hi} + \eta N_{Ai}} - (\mu_{Hi} + \kappa_{Hi})E_{Hi} + m_{ji}E_{Hj}(t) - m_{ij}E_{Hi}(t), \geq \frac{\sigma_{Vi}(t)(1-u_i)\beta_{VHi}I_{Vi}(S_{Hi}^0 - \varepsilon)}{(N_{Hi}^0 + \varepsilon) + \eta(N_{Ai}^0 + \varepsilon)} - \mu_{Hi}E_{Hi} - \kappa_{Hi}E_{Hi} + m_{ji}E_{Hj}(t) - m_{ij}E_{Hi}(t), \\ &= \left(\frac{\sigma_{Vi}(t)(1-u_i)\beta_{VHi}S_{Hi}^0}{N_{Hi}^0 + \eta N_{Ai}^0} \right) \left(\frac{1 - \frac{\varepsilon}{S_{Hi}^0}}{1 + \frac{\varepsilon(1+\eta)}{N_{Hi}^0 + \eta N_{Ai}^0}} \right) I_{Vi} - (\mu_{Hi} + \kappa_{Hi})E_{Hi} + m_{ji}E_{Hj} - m_{ij}E_{Hi}. \end{aligned} \quad (33)$$

$$E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0)) \in Y_0$$

Without loss of generality, we assume that

$$\begin{aligned} P^m(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), \\ E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0))) - M_0| < \delta, \quad \forall m \geq 0. \end{aligned} \quad (28)$$

For clinically infected humans we have:

$$\dot{I}_{Hi}(t) = \kappa_{Hi}E_{Hi} - (\mu_{Hi} + \alpha_{Hi})I_{Hi} + n_{ji}I_{Hj} - n_{ij}I_{Hi}.$$

The equations of the exposed animal populations become:

$$\begin{aligned} E'_{Ai}(t) &= \frac{\eta\sigma_{Vi}(t)\beta_{VAi}I_{Vi}S_{Ai}}{N_{Hi} + \eta N_{Ai}} - (\mu_{Ai} + \kappa_{Ai})E_{Ai} + p_{ji}(t)E_{Aj} - p_{ij}(t)E_{Ai}, \geq \frac{\eta\sigma_{Vi}(t)\beta_{VAi}I_{Vi}(S_{Ai}^0 - \varepsilon)}{(N_{Hi}^0 + \varepsilon) + \eta(N_{Ai}^0 + \varepsilon)} - (\mu_{Ai} + \kappa_{Ai})E_{Ai} + p_{ji}(t)E_{Aj}(t) - p_{ij}(t)E_{Ai}(t), \\ &= \left(\frac{\eta\sigma_{Vi}(t)\beta_{VAi}S_{Ai}^0}{N_{Hi}^0 + \eta N_{Ai}^0} \right) \left(\frac{1 - \frac{\varepsilon}{N_{Ai}^0}}{1 + \frac{\varepsilon(1+\eta)}{N_{Hi}^0 + \eta N_{Ai}^0}} \right) I_{Vi} - (\mu_{Ai} + \kappa_{Ai})E_{Ai} + p_{ji}(t)E_{Aj} - p_{ij}(t)E_{Ai}. \end{aligned}$$

Thus,

$$\begin{aligned} ||u(t, P^m(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), \\ I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0))) - u(t, M_0)|| < \varepsilon, \\ \forall t \in [0, \omega], \quad \text{and} \quad m \geq 0. \end{aligned} \quad (29)$$

Moreover, for any $t \geq 0$, we write $t = t_0 + q\omega$ with $t_0 \in [0, \omega)$ and $q =$

$\left\lceil \frac{t}{\omega} \right\rceil$, the greatest integer less than or equal to $\frac{t}{\omega}$. Then we obtain

For infectious animals we have the following outcome:

$$I'_{Ai} = \kappa_{Ai}E_{Ai} - (\mu_{Ai} + \alpha_{Ai})I_{Ai} + q_{ji}(t)I_{Aj} - q_{ij}(t)I_{Ai}.$$

The equations for the exposed vectors are given by:

$$\begin{aligned} ||u(t, (S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), R_{Hi}(0), S_{Ai}(0), \\ E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), I_{Vi}(0))) - u(t, M_0)|| = ||u(t_0, P^m(S_{Hi}(0), E_{Hi}(0), I_{Hi}(0), \\ R_{Hi}(0), S_{Ai}(0), E_{Ai}(0), I_{Ai}(0), R_{Ai}(0), S_{Vi}(0), E_{Vi}(0), \\ I_{Vi}(0))) - u(t_0, M_0)|| < \varepsilon, \end{aligned} \quad (30)$$

Table 2

Description of model parameters of system (5), indicating baseline, ranges and references.

Symbol	Baseline value	Unit	Source
N_{H1}, N_{H2}	1000	Humans	[1]
N_{A1}, N_{A2}	2000	Animals	[1]
N_{V1}, N_{V2}	6000	Tsetse flies	[1]
η	25	Unitless	[1]
τ	50	Days	[20]
b_{Ai}, b_{Hi}	$\frac{1}{15 \times 365}, \frac{1}{50 \times 365}$	Day ⁻¹	[20]
b_{Vi}^0	$\frac{1}{33}$	Day ⁻¹	[20]
μ_{Vi}^0	$\frac{1}{33}$	Day ⁻¹	[20]
μ_{Ai}, μ_{Hi}	$\frac{1}{15 \times 365}, \frac{1}{50 \times 365}$	Day ⁻¹	[20]
κ_{Vi}^0	$\frac{1}{25} \left(\frac{1}{25} - \frac{1}{30} \right)$	Day ⁻¹	[20]
κ_{Ai}, κ_{Hi}	$\frac{1}{12} \left(\frac{1}{10} - \frac{1}{14} \right)$	Day ⁻¹	[20]
σ_{Vi}^0	$\frac{1}{4} \left(\frac{1}{10} - \frac{1}{3} \right)$	Day ⁻¹	[20]
α_{Ai}, α_{Hi}	$\frac{1}{25}, \frac{1}{30}$	Day ⁻¹	[20]
γ_{Ai}, γ_{Hi}	$\frac{1}{75}, \frac{1}{90}$	Day ⁻¹	[20]
β_{VAi}, β_{VHi}	0.62	Unitless	[20]
β_{AVi}, β_{HVi}	0.01	Unitless	[20]
m_{ij}	varied	Day ⁻¹	
n_{ij}	varied	Day ⁻¹	
p_{ij}^0	varied	Day ⁻¹	
q_{ij}^0	varied	Day ⁻¹	
θ_{ij}^0	varied	Day ⁻¹	

$$\begin{aligned}
 E'_{Vi}(t) &\geq \frac{\sigma_{Vi}(t)(1-u_i)\beta_{HVi}I_{Hi}(S_{Vi}^0 - \varepsilon)}{(N_{Hi}^0 + \varepsilon) + \eta(N_{Ai}^0 + \varepsilon)} \\
 &+ \frac{\eta\sigma_{Vi}(t)\beta_{AVi}I_{Ai}(S_{Vi}^0 - \varepsilon)}{(N_{Vi}^0 + \varepsilon) + \eta(N_{Ai}^0 + \varepsilon)} - \kappa_{Vi}(t)E_{Vi} - \mu_{Vi}(t)E_{Vi} + \theta_{ji}(t)E_{Vj} - \theta(t)_{ij}E_{Vi}, \\
 &= \left[\frac{\sigma_{Vi}(t)(1-u_i)\beta_{HVi}S_{Vi}^0}{N_{Hi}^0 + \eta N_{Ai}^0} \right] \times \left[1 - \frac{1 - \frac{\varepsilon}{N_{Vi}^0}}{1 + \frac{\varepsilon(1+\eta)}{N_{Hi}^0 + \eta N_{Ai}^0}} \right] I_{Hi} \\
 &+ \left(\frac{\eta\sigma_{Vi}(t)\beta_{AVi}S_{Vi}^0}{N_{Hi}^0 + \eta N_{Ai}^0} \right) \left(1 - \frac{1 - \frac{\varepsilon}{N_{Vi}^0}}{1 + \frac{\varepsilon(1+\eta)}{N_{Hi}^0 + \eta N_{Ai}^0}} \right) I_{Ai} \\
 &- (\kappa_{Vi}(t) + \mu_{Vi}(t))E_{Vi} + \theta(t)_{ji}E_{Vj} - \theta_{ij}(t)E_{Vi}.
 \end{aligned}$$

Finally, the equations of the infectious vector becomes:

$$\dot{I}_{Vi}(t) = \kappa_{Vi}(t)E_{Vi} - \mu_{Vi}(t)I_{Vi} + \theta(t)_{ji}I_{Vi} - \theta_{ij}(t)I_{Vi}.$$

Hence we obtain

$$\frac{d}{dt} \begin{bmatrix} E_{H1} \\ I_{H1} \\ E_{A1} \\ I_{A1} \\ E_{V1} \\ I_{V1} \\ E_{H2} \\ I_{H2} \\ E_{A2} \\ I_{A2} \\ E_{V2} \\ I_{V2} \end{bmatrix} \geq [F - V - \varepsilon K] \begin{bmatrix} E_{H1} \\ I_{H1} \\ E_{A1} \\ I_{A1} \\ E_{V1} \\ I_{V1} \\ E_{H2} \\ I_{H2} \\ E_{A2} \\ I_{A2} \\ E_{V2} \\ I_{V2} \end{bmatrix}, \quad (34)$$

where

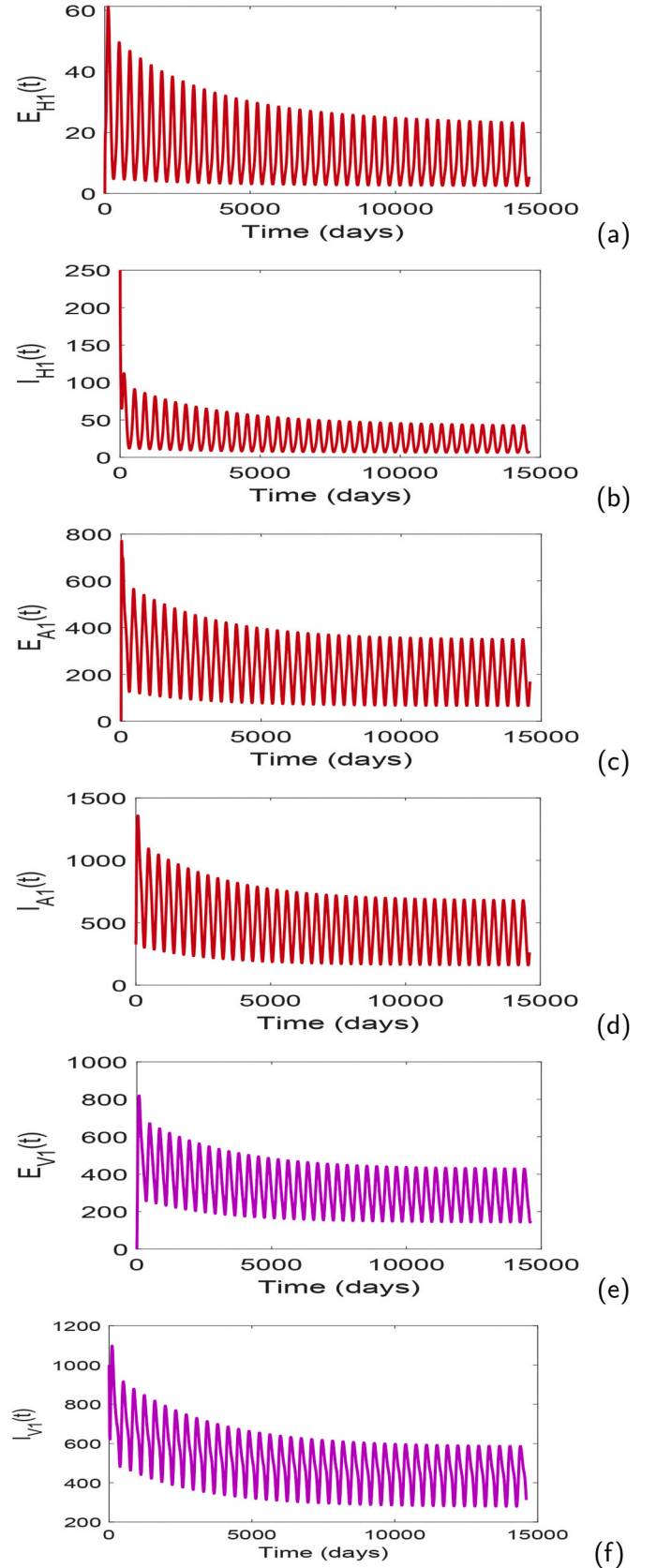


Fig. 1. Simulation solutions of model system (5) showing the effects of symmetrical movement of sleeping sickness. Model parameters used are in Table 2 and we have also set: $m_{12} = m_{21} = 0.01$, $n_{12} = n_{21} = 0.01$, $p_{12} = p_{21} = 0.1$, $q_{12} = q_{21} = 0.1$, $\theta_{12} = \theta_{21} = 0.1$, $u_1 = u_2 = 0.75$.

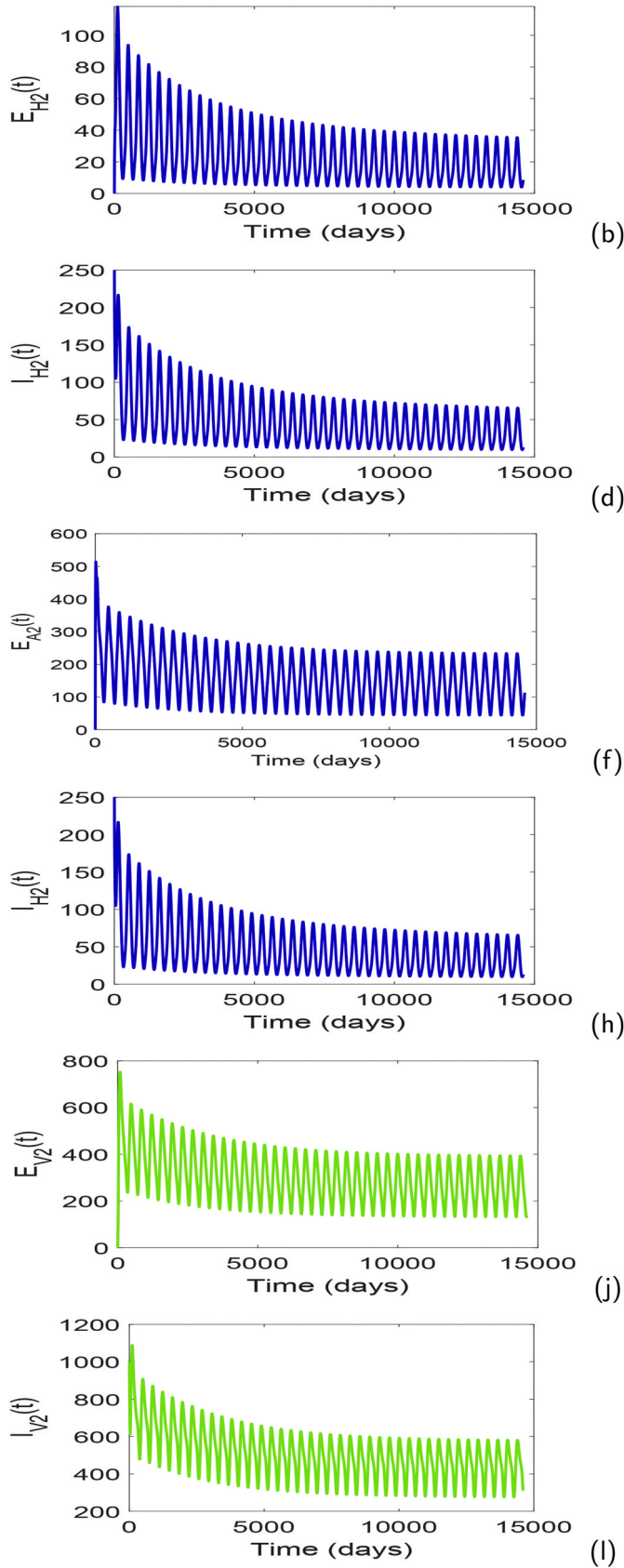


Fig. 2. Simulation solutions of model system (5) showing the effects of symmetrical movement of sleeping sickness. Model parameters used are in Table 2 and we have also set: $m_{12} = m_{21} = 0.01$, $n_{12} = n_{21} = 0.01$, $p_{12} = p_{21} = 0.1$, $q_{12} = q_{21} = 0.1$, $\theta_{12} = \theta_{21} = 0.1$, $u_1 = u_2 = 0.75$.

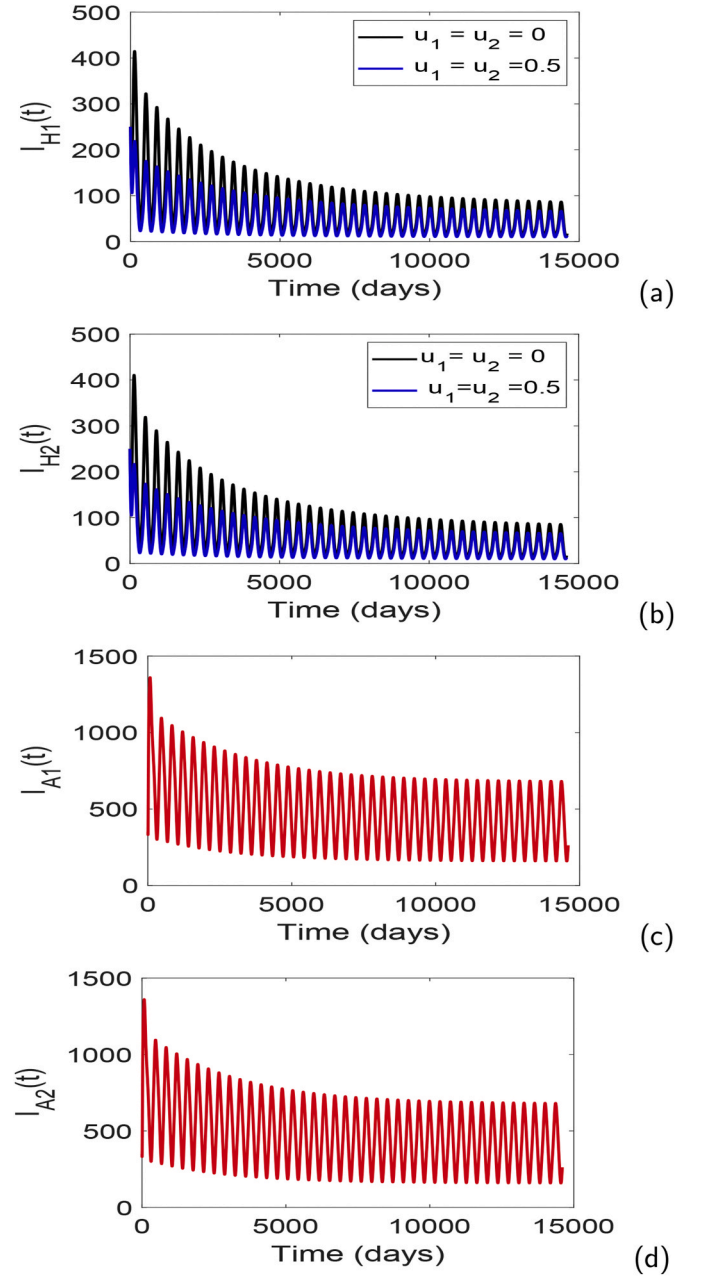


Fig. 3. Long term behavior of infected compartments for the two patches under symmetrical movement of the host and vector but low (compared to that of Fig. 1) and similar awareness level. Simulation results were obtained by setting $m_{12} = m_{21} = 0.01$, $n_{12} = n_{21} = 0.01$, $p_{12} = p_{21} = 0.1$, $q_{12} = q_{21} = 0.1$, $\theta_{12} = \theta_{21} = 0.1$, $u_1 = u_2 = 0.5$.

$$K = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & k_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & k_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & k_3 & 0 & k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k_6 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & k_7 & 0 & k_8 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (35)$$

with;

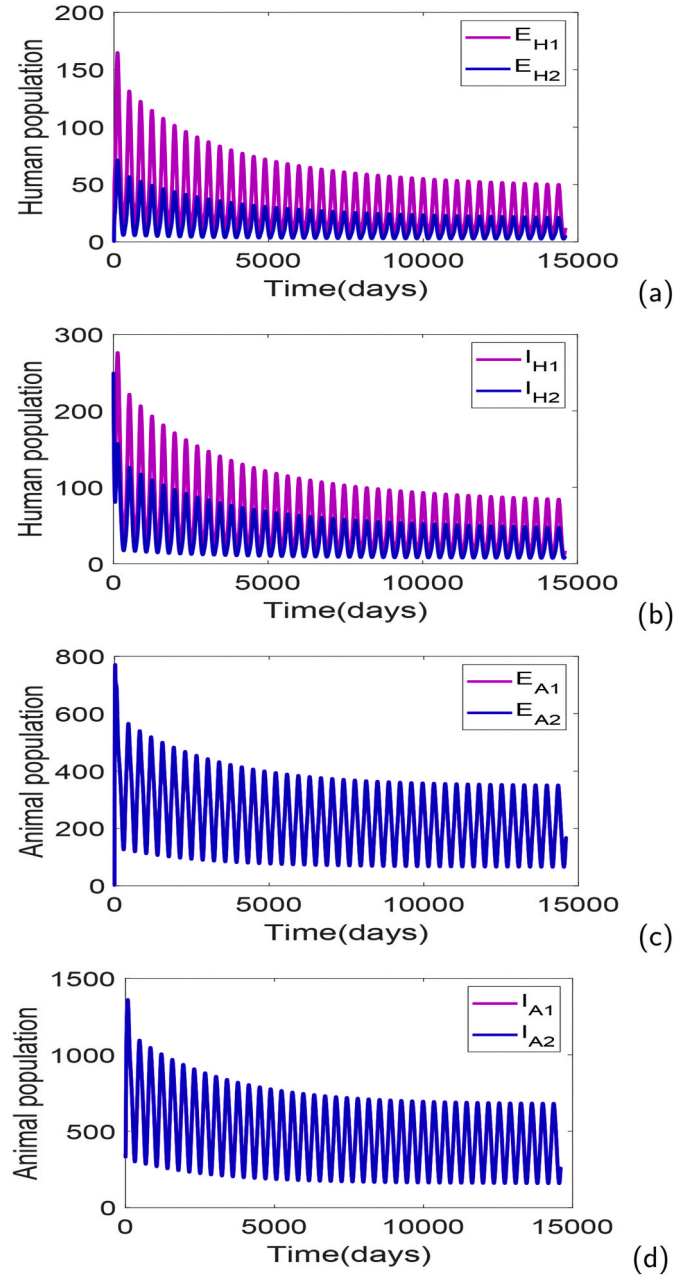


Fig. 4. Numerical simulations to demonstrate the effects of human awareness on r-HAT disease transmission. We set our controls $u_1 = 0.25$ for patch 1 and $u_2 = 0.75$ for patch 2. We also used the parameter values in Table 2 and also by setting $m_{12} = m_{21} = 0.01$, $n_{12} = n_{21} = 0.01$, $p_{12} = p_{21} = 0.1$, $q_{12} = q_{21} = 0.1$, $\theta_{12} = \theta_{21} = 0.1$. Overall, we can note that the number of new infection for the human population generated in the patch 1 is higher compared to that of the patch 2.

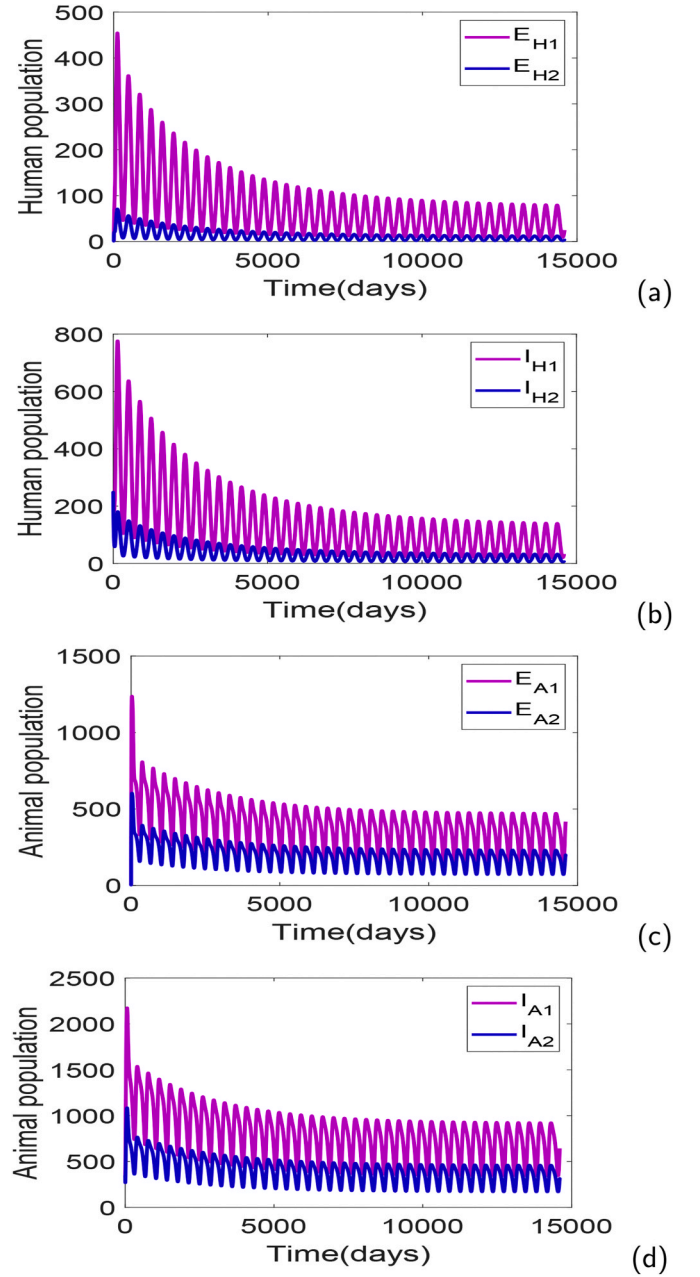


Fig. 5. Numerical simulations to demonstrate the effects of host and vector migration in the dynamics of sleeping sickness disease. In order to account for asymmetrical movement of the host and vector we set $m_{12} = 0.02$, $m_{21} = 0.06$, $n_{12} = 0.01$, $n_{21} = 0.03$, $p_{12} = 0.1$, $p_{21} = 0.2$, $q_{12} = 0.1$, $q_{21} = 0.2$, $\theta_{12} = 0.1$, $\theta_{21} = 0.13$. Thus a higher population of the host and vector is migrating from patch 2 to patch 1. In addition, we also assume similar level of awareness $u_1 = u_2 = 0.5$ for both patches.

$$k_1 = \frac{\varepsilon}{N_{H1}^0 \left(1 + \frac{\varepsilon(1+\eta)}{N_{H1}^0 + \eta N_{A1}^0}\right)}, k_2 = \frac{\varepsilon}{N_{H1}^0 \left(1 + \frac{\varepsilon(1+\eta)}{N_{H1}^0 + \eta N_{A1}^0}\right)},$$

$$k_3 = \frac{\varepsilon}{N_{V1}^0 \left(1 + \frac{\varepsilon(1+\eta)}{N_{V1}^0 + \eta N_{A1}^0}\right)}, k_4 = \frac{\varepsilon}{N_{V1}^0 \left(1 + \frac{\varepsilon(1+\eta)}{N_{V1}^0 + \eta N_{A1}^0}\right)}, k_5 = \frac{\varepsilon}{N_{H2}^0 \left(1 + \frac{\varepsilon(1+\eta)}{N_{H2}^0 + \eta N_{A2}^0}\right)},$$

$$k_6 = \frac{\varepsilon}{N_{H2}^0 \left(1 + \frac{\varepsilon(1+\eta)}{N_{H2}^0 + \eta N_{A2}^0}\right)},$$

$$k_7 = \frac{\varepsilon}{N_{V2}^0 \left(1 + \frac{\varepsilon(1+\eta)}{N_{V2}^0 + \eta N_{A2}^0}\right)}, k_8 = \frac{\varepsilon}{N_{V2}^0 \left(1 + \frac{\varepsilon(1+\eta)}{N_{V2}^0 + \eta N_{A2}^0}\right)},$$

Note that $R_0 > 1$ if and only if $\rho(\Phi_{F-V}(\omega)) > 1$. Thus, for $\varepsilon > 0$ small enough we have $\rho(\Phi_{F-V-\varepsilon,K}(\omega)) > 1$. Using Lemma 1 and the comparison principle, we immediately obtain

$$\begin{aligned} \lim_{t \rightarrow \infty} E_{Hi}(t) &= \infty, & \lim_{t \rightarrow \infty} I_{Hi}(t) &= \infty, \\ \lim_{t \rightarrow \infty} E_{Ai}(t) &= \infty, & \lim_{t \rightarrow \infty} I_{Ai}(t) &= \infty, \\ \lim_{t \rightarrow \infty} E_{Vi}(t) &= \infty, & \lim_{t \rightarrow \infty} I_{Vi}(t) &= \infty, \end{aligned} \quad (36)$$

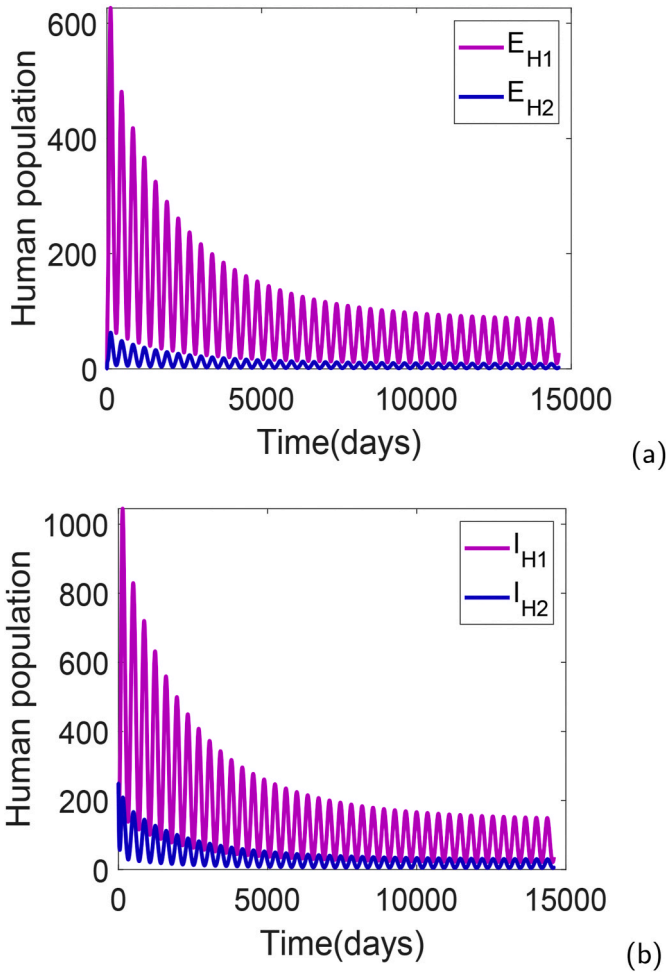


Fig. 6. Numerical simulations to demonstrate the effects of host and vector migration in the dynamics of sleeping sickness disease. In order to account for asymmetrical movement of the host and vector we set $m_{12} = 0.02$, $m_{21} = 0.06$, $n_{12} = 0.01$, $n_{21} = 0.03$, $p_{12} = 0.1$, $p_{21} = 0.2$, $q_{12} = 0.1$, $q_{21} = 0.2$, $\theta_{12} = 0.1$, $\theta_{21} = 0.13$. Thus a higher population of the host and vector is migrating from patch 2 to patch 1. In addition, we also assume similar level of awareness $u_1 = 0.25$, $u_2 = 0.75$ for both patches.

which is a contradiction.

Hence, M_0 is acyclic in M_∂ , and P is uniformly persistent with respect to $(Y_0, \partial Y_0)$, which implies the uniform persistence of the solutions to the original system [32]. Consequently, the Poincaré map P has a fixed point $(\tilde{S}_{H1}(0), \tilde{E}_{H1}(0), \tilde{I}_{H1}(0), \tilde{R}_{H1}(0), \tilde{S}_{A1}(0), \tilde{E}_{A1}(0), \tilde{I}_{A1}(0), \tilde{R}_{A1}(0), \tilde{S}_{V1}(0), \tilde{E}_{V1}(0), \tilde{I}_{V1}(0)) \in Y_0$ with $(\tilde{S}_{H1}(0) \neq 0, \tilde{S}_{A1}(0) \neq 0, \tilde{S}_{V1}(0) \neq 0)$. Thus, $(\tilde{S}_{H1}(0), \tilde{E}_{H1}(0), \tilde{I}_{H1}(0), \tilde{R}_{H1}(0), \tilde{S}_{A1}(0), \tilde{E}_{A1}(0), \tilde{I}_{A1}(0), \tilde{R}_{A1}(0), \tilde{S}_{V1}(0), \tilde{E}_{V1}(0), \tilde{I}_{V1}(0)) \in \text{Int}(\mathbb{R}_+)^{12}$ and is a positive ω -periodic solution of the system.

$$\begin{aligned} & (\tilde{S}_{H1}(t), \tilde{E}_{H1}(t), \tilde{I}_{H1}(t), \tilde{R}_{H1}(t), \tilde{S}_{A1}(t), \tilde{E}_{A1}(t), \tilde{I}_{A1}(t), \\ & \tilde{R}_{A1}(t), \tilde{S}_{V1}(t), \tilde{E}_{V1}(t), \tilde{I}_{V1}(t)) = u(t) (\tilde{S}_{H1}(0), \tilde{E}_{H1}(0), \\ & \tilde{I}_{H1}(0), \tilde{R}_{H1}(0), \tilde{S}_{A1}(0), \tilde{E}_{A1}(0), \tilde{I}_{A1}(0), \tilde{R}_{A1}(0), \\ & \tilde{S}_{V1}(0), \tilde{E}_{V1}(0), \tilde{I}_{V1}(0)) \end{aligned} \quad (37)$$

2.3. Numerical results and discussions

In order to support analytical findings, in this section model (5) will

be simulated using parameter values in Table 2 and the following assumed initial population levels. $S_{H1} = 10000$, $E_{H1} = 0$, $I_{H1} = 150$, $R_{H1} = 0$, $S_{A1} = 5000$, $E_{A1} = 0$, $I_{A1} = 50$, $R_{A1} = 0$, $S_{V1} = 20000$, $E_{V1} = 0$, $I_{V1} = 1000$. We will use the following formulae to estimate the total number of new infections generated per patch over a time period $t \in [0, t_f]$, for human and cattle populations, respectively

$$\begin{aligned} C_{H1} &= \int_0^{t_f} \left(\frac{\sigma_{V1}(t)(1-u_1)\beta_{VH1}I_{V1}(t)}{N_{H1}(t) + \eta N_{A1}(t)} \right) dt, \\ C_{A1} &= \int_0^{t_f} \left(\frac{\eta \sigma_{V1}(t)\beta_{VA1}I_{V1}(t)}{N_{H1}(t) + \eta N_{A1}(t)} \right) dt. \end{aligned} \quad (38)$$

We will explore the following scenarios:

- (a) Symmetrical movement of the host and vector
- (b) Asymmetrical movement of the host and vector

2.3.1. Case (a) symmetrical movement of the host and vector

Numerical results in Figs. 1 and 2 illustrates the solution profile of the model system (5) for infected host and vector. The results were obtained by using parameter values in Table 2. Overall, the results show that the solution profile is associated with periodic oscillations caused by change of seasonal variations. Under symmetric migration of the host and vector we can also note that the disease patterns will be the same within the two patches. From the solution profiles we can note that, at the start, the amplitude of the oscillation is very high and it decreases with time before it eventually stabilizes. Since these oscillation will not converge to the x -axis it follows that for this scenario the disease will be endemic in both patches with similar dynamics. Using formulae (38) we have found out that in each patch approximately 2.9969×10^4 sleeping sickness cases will be observed in the human population and also 2.9197×10^5 in animals.

Simulation results in Fig. 3 shows the dynamics of the disease in both patches under symmetrical movement of the host and vector as well as similar awareness levels in the two patches. Unlike in Fig. 1 where we set $u_1 = u_2 = 0.75$, here we assumed $u_1 = u_2 = 0.5$ and we can observe that a slight decrease in awareness level will have a significant impact on disease prevalence in the two patches. One can easily note that in Fig. 3 the amplitudes of the oscillations have increased significantly compared to those in Fig. 1. Therefore, one can conclude that the level of awareness has a strong influence on long-term dynamics of the disease.

Simulation results in Fig. 4 shows the effects of different awareness levels, $u_1 \neq u_2$, under the scenario of symmetrical movement. Here we have set $u_1 = 0.25$ and $u_2 = 0.75$, that is, humans from patch are considered to be high risk since their level of awareness is low that of individuals in patch 2. As one can observe, disease prevalence will be high in patch 1 than in patch 2 due to their low level of awareness. We can also note that despite high level of awareness the disease will not become extinct in patch 2 and this is because of host and vector migration. For human population in the two patches, we have noted that the following cases will be recorded in patch 1 and 2 respectively over the entire time horizon, 2.4659×10^4 and 1.5129×10^5 . Although, patch 1 will record higher cases of the disease compared to patch 2, overall, one can conclude that migration has a strongly influence on the dynamics of the disease in both patches.

2.3.2. Case (b) asymmetrical movement of the host and vector

In this scenario, we will explore the implications of unequal rates of movement for the host and vector. Precisely, we will assumed a higher population of the host and vector is migrating from patch 2 to patch 1. Thus we set $m_{12} = 0.02$, $m_{21} = 0.06$, $n_{12} = 0.01$, $n_{21} = 0.03$, $p_{12} = 0.1$, $p_{21} = 0.2$, $q_{12} = 0.1$, $q_{21} = 0.2$, $\theta_{12} = 0.1$, $\theta_{21} = 0.13$. Simulation results in Fig. 5 shows the dynamics of the disease in two patches when movement of the host and vector is asymmetric and also when the level of awareness is the same for the two patches, that is, $u_1 = u_2 = 0.5$. As we can observe, disease prevalence will be higher in the receiving patch

(patch 1) clearly demonstrating that migration the host and vector has a strongly influence on the spread of sleeping sickness. For the entire time horizon, $t = 15000$, the following sleeping sickness cases will be recorded in patch 1 and 2, respectively, among the human population, 2.4067×10^4 and 8.1645×10^4 . Thus, sleeping sickness cases will be approximately 81,645 higher in patch 1 than in patch 2.

In Fig. 6 we examine the effects of different levels of awareness $u_1 \neq u_2$ under the asymmetric scenario. We set $u_1 = 0.25$ and 0.75 . As we can observe, sleeping sickness prevalence will significantly decrease in patch 2 due higher emigration and awareness that in patch 1. For the entire time horizon, $t = 15000$, approximately 1.0482×10^5 sleeping sickness cases for the human population will be recorded in patch 1 compared to 1.0449×10^4 that will be observed in patch 2. Thus, patch 1 will account for approximately 94,371 higher than those observed in patch 2.

2.4. Concluding remarks

In this paper a two-patch non-autonomous sleeping sickness model has been proposed and analyzed. The framework incorporates seasonal variations, host and vector movement as well vector-bias. As suggested in prior studies, the proposed model includes two host, humans and animals. To account for sleeping sickness preventative strategies we incorporated a parameter in our model that represent awareness level of the human population in each patch. This parameter has been introduced such that when awareness is 100% then no humans will be infected with the disease. Changes in seasonal patterns is assumed to impact animals and vector movement. We computed the basic reproduction number, $\mathcal{R}_0$, of the model and established the threshold result: when $\mathcal{R}_0 < 1$, the disease-free equilibrium is globally stable; when $\mathcal{R}_0 > 1$, the disease is uniformly persistent. We investigated the effects of symmetric and asymmetric movement of the host and vector under different levels of awareness. It has been established that if movement of the host and vector is symmetric between the two patch and also if the two patches have similar level of awareness the disease patterns will be similar in the two patches. However, if movement is symmetric and the level of awareness differ then the patch with low awareness level will be associated with more infections. Under asymmetrical conditions, we have shown that a patch with higher emigration rates is likely to have less number of infections over time compared to the one with low emigration of the host and vector. Our results has reaffirmed earlier findings that migration of the host and vector strongly influences sleeping sickness dynamics, hence it is essential for policy makers to include spatial and seasonal heterogeneities in the design of control strategies for sleeping sickness.

A potential limitation of the present paper is that we could not validate our proposed framework with sleeping sickness data due to its unavailability, however, we plan to do so in our future extensions to this work once we managed to find appropriate datasets. In addition, we have used ordinary differential equations (ODEs) approach to model host and vector movement. We also plan to extend this work to utilize partial differential equations (PDEs), for example, by adding diffusion terms to an ODE model so as to represent the movement and dispersal of the host and vector. Seasonal variation can be similarly incorporated by considering periodic model parameters, resulting in a periodic PDE system. Such a periodic PDE model can be possibly analyzed using techniques recently proposed by Zhao and co-workers [33]. Comparison of the results obtained from these two different approach would certainly improve our existing knowledge on understanding the implications of seasonality, host and vector migration on sleeping sickness transmission and control. In addition, due to unavailability of reliable dataset, we were unable to validate the proposed sleeping sickness model. We hope in future we will be able to finding appropriate dataset and validate our work.

Availability of data and materials

All data have been included in the manuscript.

Consent for publication

No personal information is used in this study, thus Not Applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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